that the likelihood for double strand breaks of the DNA is greater and the cell repair mechanism less efficient when the DNA is in the condensed chromosome state as compared to when it is in the chromatin state (S-state). In Figure 15.3, the response function for mitosis seems to be linear while the "late S" function is curved. Extrapolation of the straight part of the S-curve extends it to the zero dose line at ~100; this number is referred to as the *extrapolation number*, *N*. Its significance is not quite clear. The typical values for human cells and X- and γ-radiation are 2−10. The value of the extrapolated line at 100% cell survival (or $N = 1$) is referred to as the "*quasi-threshold dose*", D_q. The "shoulder" in the dose-response curve is typical for X- and γ-irradiation of human cells while neutrons and alpha particles hardly produce any shoulder. Also, a shoulder does not form when very simple organisms, such as viruses, are irradiated by X- and γ-rays. A simple straight dose-response relation (the mitosis line) is taken as support of a *single-cell hit killing*, which is purely random. We have already concluded that n and α radiations are likely to severely damage DNA, particulary during the G_2 and mitosis stages.

The existence of a shoulder is taken as support of a *multi-target model*: the DNA must be damaged at several points by X- or γ-rays in order to cause the cell to die because of the efficient repair mechanism for single-strand breaks.

The slope of the straight part of the line allows calculation of the *mean lethal dose*, D_0, which is the dose required to inactivate the fraction $1 - e^{-1}$ (i.e. 63%) of the cells. Designating the surviving fraction as f_S gives

$$f_S = e^{-D/D_0} \tag{15.3}$$

for the straight line (mitosis). The curve can be described by

$$f_S = 1 - \left(1 - e^{D/D_0}\right)^N \tag{15.4}$$

15.5. FURTHER REGULARITIES AT LARGE DOSES

Figure 15.4 shows the number of dicentric chromosome abberations observed in a cell population as a function of neutron or gamma/X-ray energy. *Dicentric abberations* (Fig. 15.1.D) seem to be the most consistent index of radiation damage and represents about 60% of all observable unstable abberations following acute irradiation. Their background frequency is very low (about 1 in 1000 lymphocyte cells). From Figure 15.4, it is seen that the lower the radiation particle energy is, the larger the damage (of course, until the energy has decreased to such low levels that its ionization effects begin to drop abruptly, see Figure 7.7). This is attributed to the increasing ionization density with decreasing particle energy, as described in chapter 7; see also Tables 15.2 and 15.4.

Figure 15.4 also shows that the damage increases with increasing *dose rate*. This is a general and significant phenomenon which is probably related to the cell (DNA) repair

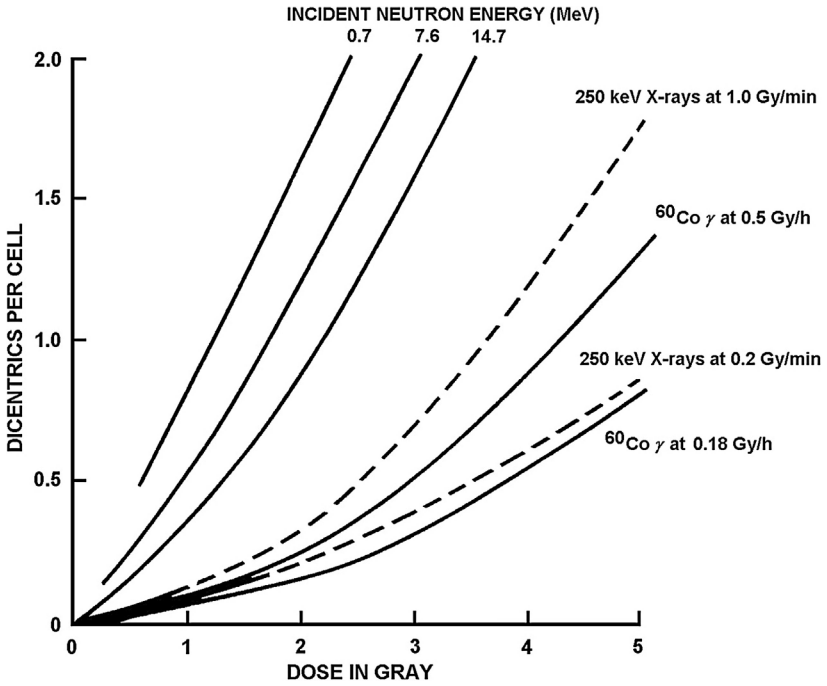


Figure 15.4 Dose response curves of dicentric abberations yields for several qualities of radiation *(Lloyd and Purrot 1981, from Nias.)*

mechanism. At high dose rates, the repair mechanism may become saturated. If a large number of cells are damaged almost simultaneously, the tissue may cease to function. This is seen in the nerve cells at large doses and dose rates.

In animal experiments, it has been found that a higher dose rate also produces earlier cancerogenesis (i.e. shorter latency) and more severe forms of the tumor (higher malignancy).

Figure 15.5.A illustrates the effect of *dose fractionation* on cell survival at high doses and dose rates. After each succeeding dose fraction, the curve exhibits the same shoulder and slope, indicating that within a given period part or all of the damage is repaired.

Figure 15.5.B illustrates a matrix effect during irradiation in air and in an oxygen deficient (*hypoxic*) system. OER, the oxygen enhancement ratio, is the relative dose increase needed to produce the same biological effect in the hypoxic as the oxic case. For X-rays and γ, cell survival increases when oxygen is reduced; the addition of oxidants has the opposite effect. This indicates that it is the oxidative radiolysis products of water which are most damaging to the cell. However, no oxygen effect is seen for α radiation, thereby supporting the model of double strand breaks by direct hits of the α's.

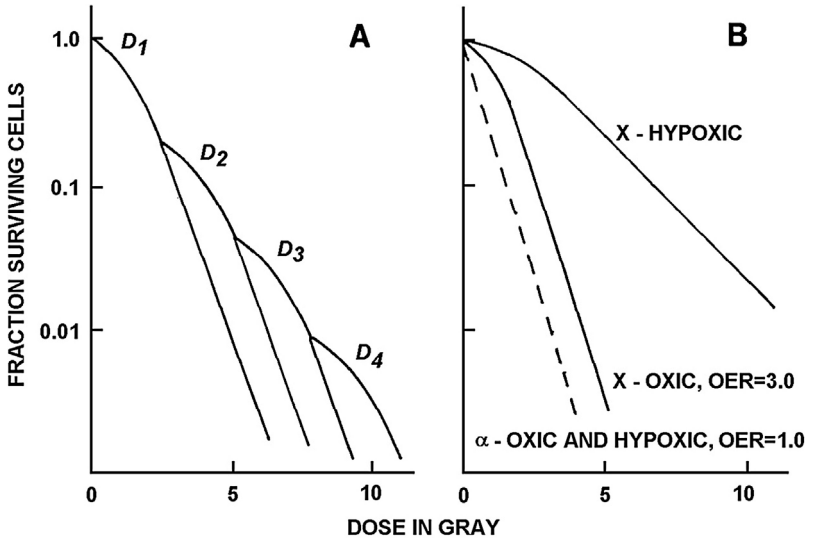


Figure 15.5 Fraction of surviving mammalian cells as a function of dose. A) Effect of dose fractionation. B) Irradiation by X-rays and by α-particles under aerated and hypoxic conditions.

This result suggests that the addition of a reductant would reduce the radiation effect which is also observed. For example, 20 Gy is required to kill 80% of a cell population in the presence of cysteine (or cysteamine), while in the reference system without cysteine the same effect is caused by only 5 Gy. Cysteine is oxidized to cystine.

Several other *radiation protection agents* are known. These compounds are typically amino thiols, similar to the natural amino acid cysteamine. They probably function as scavengers for the products of water radiolysis. Their effectiveness is evaluated by determination of the *dose reduction factor* (DRF), which is the ratio of $LD_{50/30}$ for protected and unprotected animals. Because of their chemical toxicity, many can only be administered in small doses.

No evidence has been found for cells or of higher organisms supporting the development of long-lasting *radiation resistance*. Certain bacteria have been shown to develop a seeming resistance to radiation after receiving small radiation doses over a long period of time. However, this is possibly due to the formation of mutated organisms with a different sensitivity to radiation than the original ones.

15.6. EPIDEMIOLOGICAL OBSERVATIONS OF EFFECTS AT LARGE RADIATION DOSES

We may distinguish between accidental (e.g. Japan and Chernobyl) and deliberate (as given in radiotherapy) exposures to large radiation doses. The former are said to be